
Specialist Physical AI Need Not Be Opaque A Fully Tensor-Decomposable VLA Policy

Protocol-aligned control and exact weight analysis

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Abstract

1 Tensor decomposition turns a large table of learned interactions into simpler low-
2 rank factors that can be ranked and inspected. Conventional robot policies contain
3 operations that block this direct weight-level view. We introduce χ -VLA, a ra-
4 tional vision-language-action model family with a 19.2M convolutional policy
5 and a 20.1M ViT policy, both built only from bilinear, linear, and rational learned
6 operations. Their weights can therefore be reorganized into explicit interaction
7 tensors rather than treated only as an opaque checkpoint. A runtime operator
8 audit of the seed-0 20.1M ViT checkpoint and local numerical reconstructions
9 verify its operator-level form. Across three independently trained 20.1M ViT poli-
10 cies, closed-loop LIBERO-Object success is $85.3\% \pm 4.6\%$ over 500 canonical
11 trials per checkpoint. A fixed elementwise prediction-mean ensemble reaches
12 $467/500 = 93.4\%$ at the cost of three forward passes. Across the three check-
13 points, a prospectively frozen paired test on 100 unique familiar-task states finds
14 $247/300 = 82.3\%$ correct-instruction success, versus $72/300 = 24.0\%$ with a
15 co-present control instruction and $78/300 = 26.0\%$ with an empty instruction. The
16 paired gaps are positive for every checkpoint aggregate and every task aggregate. A
17 separate convolutional instantiation passes independent 4,608-case joint-attention,
18 384-case joint-FFN, and 384-case complete joint-block certificates. Their respec-
19 tive worst relative errors are 6.36×10^{-16} , 1.39×10^{-15} , and 1.5024×10^{-7} .
20 Exact-attention records naming those same ViT checkpoint artifacts numerically
21 replay every attention module on one fixed corrected cached input per checkpoint,
22 with worst relative error 2.016119×10^{-7} . Across 48 fixed inputs, a sequential
23 replay rebuilds all 12 learned transformer blocks to the action output at worst
24 relative error 3.9822×10^{-7} against a frozen 10^{-3} gate. In a seed-0 ViT check-
25 point, exact weight-derived subspaces beat equal-rank random controls in all 36
26 primary block-head tests. Separately, prospectively fixed rank-96 visual projectors
27 retain 95.2–98.2% of baseline successes on tasks absent from corrected projector
28 construction, while three equal-rank random projectors retain only 3.3–13.6%. The
29 result is a capable and structurally inspectable specialist family with exact access
30 to trained attention coefficients and bounded familiar-task instruction necessity.
31 It does not establish counterfactual goal following, zero-shot language use, or an
32 input-general compact symbolic decomposition of the full policy.

1 Introduction

Modern robot policies can achieve high success while giving little indication of why an action was chosen. Their checkpoints contain millions of weights, but those weights do not directly show which image regions, words, and robot-state coordinates interact. Post-hoc probes can find correlated activations, yet a correlation is not automatically a mechanism or a safe editing target.

This is not a zero-shot policy. It is trained from scratch on one specialist suite. Its role in the workshop’s question is to establish an auditable specialist control against which future zero-shot and transfer claims can be evaluated.

A tensor is simply a multidimensional table. Tensor decomposition rewrites that table as a sum of low-rank factors, analogous to finding dominant directions in a matrix. If every learned block uses additions, multiplications, and ratios of polynomials, the full policy retains an explicit interaction table defined by its weights. Its decomposed factors then provide candidate control mechanisms that can be compared across layers and tested by intervention [3, 4]. This is what we mean by a fully tensor-decomposable policy. It is an operator-closure claim, not a claim that we materialize one compact end-to-end coefficient tensor. Its degree and representation size grow rapidly with depth. For example, one coefficient can connect an image coordinate, an instruction coordinate, and an action coordinate. Factoring the coefficient table ranks high-energy cross-modal interactions directly from the checkpoint without first fitting an activation probe. Those factors are candidate structure, not automatically human-readable concepts or causal explanations.

The restriction is nontrivial in robotics. Softmax, ordinary activations, and square-root normalization break the algebraic form, while continuous actions require precise closed-loop outputs. The potential payoff is also concrete: a weight-derived mechanism could help diagnose whether a policy follows language, keys on a visual shortcut, or uses a fragile control feature.

We study the narrower but testable question:

Can a specialist VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while supporting exact weight analysis?

We test capability and structural analysis in two fully rational encoder instantiations. The same three ViT checkpoints establish closed-loop capability and support the exact attention audit. A separate convolutional instantiation tests whether structural certificates transfer across visual encoders.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate three fully rational 20.1M ViT policies on 1,500 canonical LIBERO-Object trials. They reach $85.3\% \pm 4.6\%$, while a fixed prediction-mean ensemble reaches $467/500 = 93.4\%$.
3. We mechanically audit the seed-0 20.1M ViT checkpoint and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor. In a separate closed-loop audit, learned rank-96 visual projectors retain 95.2–98.2% of baseline successes, versus 3.3–13.6% for random projectors.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive replaces a familiar transformer component while keeping explicit coefficients. Linear maps contribute first-order terms, while multiplying learned projections creates higher-order interactions that remain determined by the weights.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The inner sum is a table indexed by one output and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with interaction coefficients determined by the weights.

Rational per-instance normalization. RMS normalization contains an input-dependent square root. Let $m(x) = d^{-1} \sum_{j=1}^d x_j^2 + \epsilon$, and let $s_0 > 0$ be the frozen running mean-square scale. We deploy a fixed rational approximation $r(v) = P(v)/Q(v)$ to $v^{-1/2}$ over $v \in [0.1, 10]$:

$$v = \frac{m(x)}{s_0}, \quad \text{RNorm}(x) = \frac{x}{\sqrt{s_0}} r(v). \quad (3)$$

Here rational means a ratio of polynomials. Projective coordinates carry each activation as a numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific rescaling without leaving the rational tensor representation. Training and deployment use r itself. The exactness claims below therefore concern faithful evaluation of the deployed rational computation, not uniform agreement with RMSNorm. Appendix A.4 certifies pole-free denominators and deployed arithmetic fidelity while retaining that boundary.

2.2 Architecture and training

Both encoder variants follow the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. They use a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. Both share a width-384, eight-block, twelve-head joint backbone and a linear head that maps each action-query state to seven continuous action dimensions. The 20.1M ViT uses four bilinear-attention vision blocks and has 20,137,352 learned parameters. It supplies the closed-loop capability, runtime operator census, exact-attention, and reduced-Gram results. The 19.2M convolutional variant instead uses three stride-2 bilinear convolution stages that produce an 8×8 token grid. Three separate convolutional files receive the joint-attention and rational-normalizer certificates reported in Appendix A.4.

Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the seed-0 20.1M ViT checkpoint

Architecture definitions can differ from deployed code. We therefore audit the deployed seed-0 20.1M ViT checkpoint at runtime and reconstruct representative operations from saved weights.

The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN reconstruction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed module’s float32 statistic. The audit certifies operator membership and local reconstruction, as summarized in Appendix Figure 5. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with the following fixed protocol:

- 50 canonical trials per task
- a 280-step cap
- three independently trained ViT checkpoints
- 500 rollouts per checkpoint and 1,500 rollouts in total

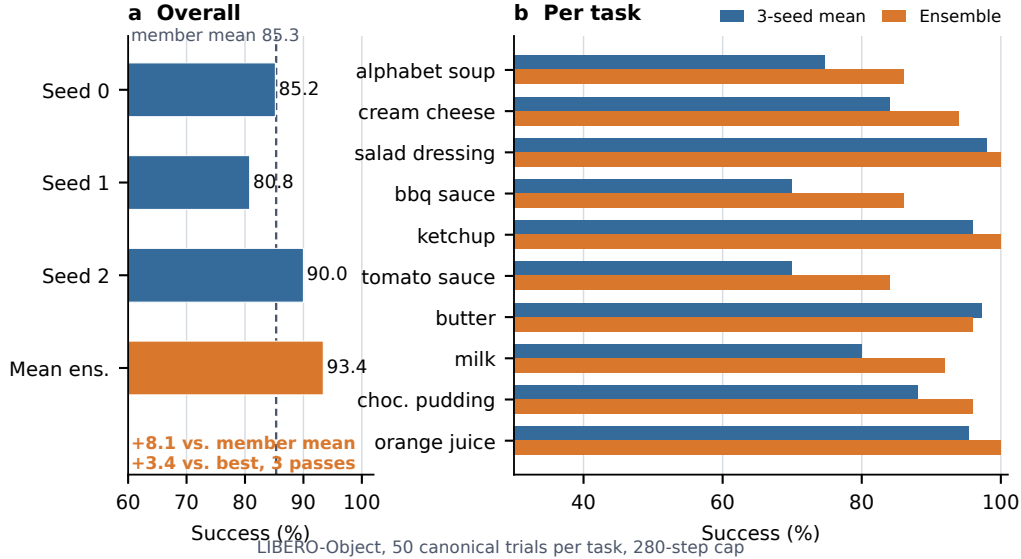


Figure 1: **Verified ViT capability.** Left: three independently trained checkpoints and their fixed elementwise prediction-mean ensemble. Right: per-task three-checkpoint mean and ensemble success. The ensemble uses three forward passes.

3.2 Results

Method	Parameters	Inference	Success
χ -VLA, rational ViT	20.1M	1 pass	$85.3\% \pm 4.6\%$
χ -VLA, prediction-mean ensemble	$3 \times 20.1M$	3 passes	93.4%

Table 1: **Closed-loop LIBERO-Object success.** Each ViT checkpoint receives 500 canonical trials. The single-model row is the mean $\pm$ sample standard deviation across three checkpoints. The ensemble is one fixed composite evaluated on the same 500 states.

131 The three checkpoints reach $426/500 = 85.2\%$, $404/500 = 80.8\%$, and $450/500 = 90.0\%$. Ele-
 132 mentwise averaging of their predicted action chunks reaches $467/500 = 93.4\%$. This is 8.1 points
 133 above the member mean and 3.4 points above the strongest member. The ensemble requires three
 134 complete forward passes and is not one 20.1M-parameter policy. Every episode-level record, task
 135 index, and protocol field is retained. The artifact manifest binds each immutable result file and
 136 recorded checkpoint basename to a checkpoint hash verified separately. Member and ensemble shards
 137 span different recorded GPU families and predate explicit matrix-precision fields, so the arithmetic
 138 gaps are descriptive rather than a matched-hardware ensemble ablation.

139 3.3 What this result does and does not establish

140 The result shows that the rational architecture supports high closed-loop specialist capability. It is an
 141 existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization
 142 or state of the art.

143 3.4 Familiar-task instruction necessity

144 Before observing these outcomes, we fixed a paired full-horizon test on 100 unique canonical states
 145 across the same three checkpoints. Correct instructions succeed on $247/300 = 82.3\%$ of checkpoint-
 146 state pairs, compared with $72/300 = 24.0\%$ for an observation-verified co-present control instruction
 147 and $78/300 = 26.0\%$ for an empty instruction. Both gaps are positive for every checkpoint aggregate
 148 and every task aggregate. The exact tests and a state-cluster bootstrap also pass their frozen gates.
 149 Appendix A.1 gives the protocol, statistics, and plot. Every tested task and instruction appeared during
 150 training, so this is instruction necessity for familiar tasks, not zero-shot language generalization or
 151 counterfactual goal following.

152 4 Exact weight-derived structure through attention

153 4.1 Exact layerwise construction

154 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives
 155 a coefficient table whose entries measure how combinations of query, key, and value coordinates
 156 contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-
 157 token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to
 158 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

159 The deployed attention also normalizes each query and key with Equation 3. Each normalization
 160 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
 161 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
 162 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
 163 10^{-16} . On one fixed corrected cached input per checkpoint, we numerically replay all four vision
 164 and eight joint attention modules in each of three records that name the same ViT checkpoint artifacts
 165 as the capability results. This covers 36 module-input cases and 384 architectural heads. The worst
 166 relative ℓ_2 error is 2.016119×10^{-7} and the worst absolute difference is 2.44×10^{-5} . This residual
 167 comes from the deployed implementation’s intentional float32 statistic, while the reconstruction itself
 168 uses the learned coefficients. The algebraic identity is input-general. The one-input-per-checkpoint
 169 records provide a layerwise numerical replay audit, not an exact end-to-end policy decomposition.

170 We separately certify three convolutional checkpoints from the same rational architecture family.
 171 Their vision stacks contain no attention, so we independently rebuild all eight joint modules and all
 172 twelve heads per module from the saved weights. Sixteen deterministically selected cache inputs
 173 per checkpoint yield 4,608 head-input cases. Every output is finite and the worst relative ℓ_2 error is
 174 6.36×10^{-16} , below the frozen 10^{-6} gate. On the same inputs, independent CP-factor evaluation
 175 covers all 384 joint-FFN module-input cases at worst relative error 1.3871×10^{-15} against a 10^{-10}
 176 gate. Independent complete-block equations cover all 384 block-input cases at worst relative error
 177 1.5024×10^{-7} against a 10^{-4} gate. The complete-block comparison crosses a deployed PyTorch
 178 float32 block call and a NumPy float64 reconstruction. It is input-conditioned and per-block, not a
 179 sequential end-to-end reconstruction of the joint transformer or policy.

A frozen ViT audit partitions joint attention into vision, instruction, robot-state, and action-query sources. Across 128 task-balanced inputs per checkpoint, all 36,864 head-input cases are finite, with worst reconstruction error 5.0741×10^{-7} against a 10^{-6} gate. Mean projected coherent-energy shares are 67.7% vision, 19.2% state, 7.9% action query, and 5.1% instruction. These shares and prompt permutations are descriptive, not causal importance, grounding, or closed-loop evidence.

On the same checkpoints, a frozen sequential NumPy float64 replay starts from exact prepared tensors and rebuilds the learned patch projection, four vision blocks, multimodal inputs, eight joint blocks, final RationalNorm, and linear head. Over 48 inputs and 816 stages, its worst final-action relative ℓ_2 error is 3.9822×10^{-7} against a frozen 10^{-3} gate, and every manual PyTorch traversal equals an unmodified call bitwise. This fixed-input numerical certificate excludes preprocessing, unnormalization, execution, simulation, and success. It is not a compact symbolic or input-general proof.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors rank important directions without constructing the full table. It matches brute-force materialization to 2.13×10^{-14} at toy scale.

4.2 Trained policy result

The reconstruction certificate above spans three ViT checkpoints. The following subspace-ranking experiment is a descriptive analysis of the seed-0 ViT checkpoint. We apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached activations are used only afterward to score how well it preserves the layer’s computation.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of 36 block-head pairs favor the exact subspace, with median ratios 1.028, 1.171, and 2.441. At rank 128, the mean is 3.033 and the maximum is 10.135. These are architectural units within one checkpoint, not independent model-level replicates.

This extends exact weight-only construction through individual attention layers. It is not an exact compact symbolic end-to-end decomposition of the complete policy. Symbolic composition across all eight blocks still faces rapid degree and representation growth.

A rank-96 action-Jacobian projector fixed from official Object tasks 0–3 retains 259/272 full-policy successes on tasks 8–9 across three ViT checkpoints. An activation-energy projector retains 267/272, while three equal-rank random projectors retain 37/272, 9/272, and 22/272. All ten tasks appeared during policy training, but tasks 8–9 are absent from corrected projector construction. This supports robustness to a structured 75% channel reduction, not zero-shot transfer, unique action-Jacobian specificity, or a causal link to the exact weight factors. Appendix A.3 reports paired controls and stability intervals.

5 Related work and limitations

Related work. OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative weights can expose low-rank structure and support subsequent controls [3–5]. Recent work also steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a policy designed as a rational tensor program. Tensor and polynomial networks provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success can coexist with weak language use [9–11].

Limitations and conclusion. The benchmark is one specialist suite. Exact layer equations and a fixed-input learned-forward replay reach the action output, but neither is a compact symbolic or input-general proof. The paired audit establishes familiar-task instruction necessity, not counterfactual goal following, zero-shot language use, or broad grounding. Within that scope, rational restrictions support strong control and trained-weight certificates. We do not claim broad generalization or coefficient editing. SHA-pinned artifacts release and verify every reported raw decision.

References

- [1] M. J. Kim et al. *OpenVLA: An Open-Source Vision-Language-Action Model*. CoRL, 2024.
- [2] M. J. Kim, C. Finn, and P. Liang. *Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success*. 2025.
- [3] A. Rigg, T. Dooms, M. T. Pearce, J. Oramas, and L. Sharkey. *Bilinear MLPs Enable Weight-Based Mechanistic Interpretability*. ICLR, 2025.
- [4] T. Dooms, W. Gauderis, G. A. Wiggins, and J. Oramas. *Compositionality Unlocks Deep Interpretable Models*. 2025.
- [5] F. Oozeer, Y. Eriskin, A. Rigg, J. Oramas, and L. Sharkey. *Bilinear Convolution Decomposition for Causal RL Interpretability*. 2024.
- [6] B. Häon, K. Stocking, I. Chuang, and C. Tomlin. *Mechanistic Interpretability for Steering Vision-Language-Action Models*. 2025.
- [7] S. Marks et al. *Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models*. ICLR, 2025.
- [8] M. Kekić et al. *Learning Nonlinear Causal Reductions to Explain Reinforcement Learning Policies*. ICLR, 2026.
- [9] X. Zhou et al. *LIBERO-PRO: Towards Robust and Fair Evaluation of Vision-Language-Action Models Beyond Memorization*. 2025.
- [10] Y. Fang et al. *When Vision Overrides Language: Evaluating and Mitigating Counterfactual Failures in VLAs*. 2026.
- [11] C. Kim et al. *LIBERO-Para: A Diagnostic Benchmark and Metrics for Paraphrase Robustness in VLA Models*. 2026.
- [12] T. Kolda and B. Bader. *Tensor Decompositions and Applications*. SIAM Review, 2009.
- [13] E. Stoudenmire and D. Schwab. *Supervised Learning with Tensor Networks*. NeurIPS, 2016.
- [14] G. Chrysos et al. *Deep Polynomial Neural Networks*. CVPR and TPAMI, 2020 to 2021.

A Supporting results and decisions

A.1 Prospectively frozen closed-loop instruction necessity

High task success alone does not show that a policy uses its instruction. Before these outcomes were observed, we fixed a paired test on canonical episodes 40–49 for every task and the same three checkpoints. Each checkpoint-state pair was rolled out for the full 280-step horizon under the correct familiar instruction, one deterministic outcome-independent co-present control instruction, and an empty instruction. All conditions shared the exact initial physical input and their order was chosen by a frozen hash. Co-presence was verified from simulator observation fields and does not guarantee unoccluded camera visibility. The empty string encodes tokenizer BOS plus 31 unmasked PAD tokens, together with the model’s learned common BOS.

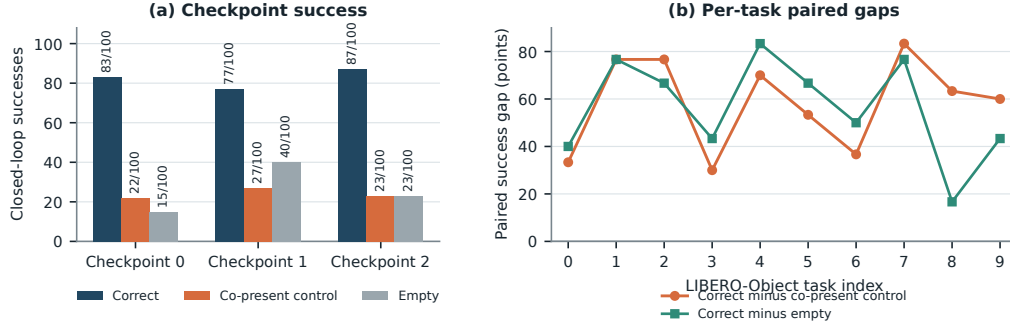


Figure 2: **Closed-loop instruction necessity on familiar tasks.** Left: exact successes among 100 paired canonical states per checkpoint. Right: correct-instruction success minus each control, pooling the three checkpoint outcomes for each task. All ten task gaps are positive for both controls at the task-aggregate level.

Across 100 unique canonical states and three checkpoints, the correct instruction succeeds on 247/300 = 82.3% of checkpoint-state pairs. The co-present control succeeds on 72/300 = 24.0% and the empty instruction on 78/300 = 26.0%, giving paired gaps of 58.3 and 56.3 points. Correct-instruction success is 83/100, 77/100, and 87/100 by checkpoint, and both paired gaps are positive for every checkpoint aggregate. All ten pooled task aggregates are strictly positive for both controls. One-sided exact McNemar tests give $p = 2.02 \times 10^{-45}$ and $p = 1.52 \times 10^{-44}$. These pooled tests use 300 checkpoint-state pairs and are not cluster-robust inference. A frozen 20,000-draw task-stratified state-cluster bootstrap resamples 100 unique task-episode states while retaining all three checkpoint outcomes. Its 95% intervals are [51.0, 65.7] and [50.0, 62.3] points. Every prospectively frozen gate passes.

This result establishes instruction necessity for familiar LIBERO-Object prompts, objects, scenes, tasks, and checkpoints. It does not show that the policy completes a counterfactual prompted goal, nor does it establish paraphrase robustness, unseen-object or compositional grounding, cross-suite generalization, physical transfer, or a benefit unique to tensor decomposability.

For WRL, the specialist uses familiar task language, but every tested task and instruction appeared during training. The result is therefore not zero-shot language generalization.

A.2 Expanded attention screen

Within the same seed-0 ViT checkpoint, the corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

A.3 Prospectively fixed-rank visual bottleneck

The behavioral intervention acts at the width-384 visual-to-joint interface and keeps rank 96. Projector construction uses official Object tasks 0–3. Rank 96 was fixed before any corrected-basis outcome was observed, and the frozen evaluation uses tasks 8–9. Across three checkpoints and 300 canonical episodes, the full policy succeeds on 272 trials. The action-Jacobian projector retains 259/272 = 95.2% of those successes, with a paired checkpoint-task-episode resampling interval of 91.2–98.6%. The activation-energy projector retains 267/272 = 98.2%, with an interval of 95.8–100.0%. Three fixed equal-rank random projectors retain 37/272 = 13.6%, 9/272 = 3.3%, and 22/272 = 8.1%.

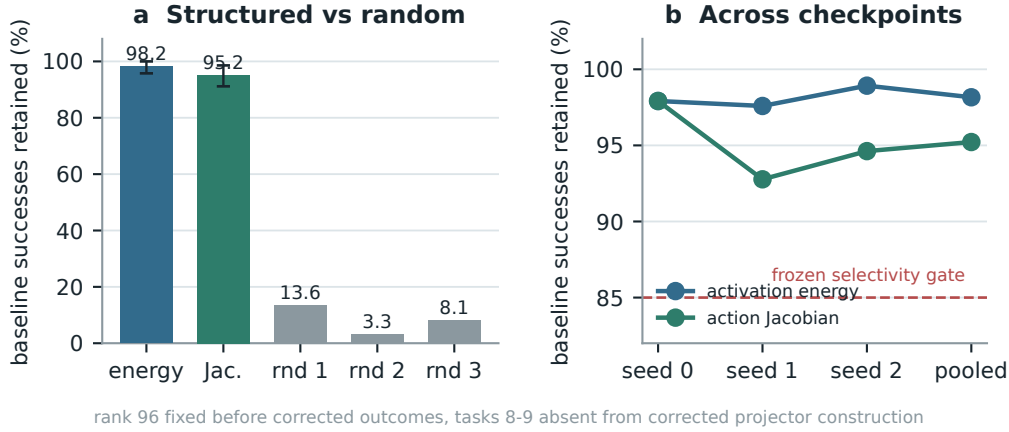


Figure 3: **Corrected closed-loop visual bottleneck.** Both learned rank-96 projectors preserve nearly all baseline successes across three checkpoints, while equal-rank random projectors do not. Error bars in panel (a) are deterministic paired cluster-resampling intervals. Tasks 8–9 are absent from corrected projector construction, but not from policy training. This is not a clean rank-selection-holdout result.

The action-Jacobian basis does not outperform the matched activation-energy basis. The stronger action-specificity gate therefore does not pass. The supported claim is structured visual sufficiency relative to random projectors. This is not zero-shot evidence, and the data-derived intervention remains separate from the exact weight-only attention decomposition.

A.4 Numerical scope of rational conversion

The deployed degree- $[2/2]$ least-squares rational fit is exact as a rational computation even though it approximates RMS normalization. Its numerator has coefficients (5.32997, 7.53552, 0.327274) and the denominator has coefficients (1, 9.64827, 2.60664). Every denominator coefficient is positive, which certifies $Q(v) > 0$ for all $v \geq 0$. We instrument all 53 RationalNorm module instances, each invoked once per forward pass, on 4,096 deterministic inputs for each of three convolutional checkpoints. Across 544,542,720 activation rows, the minimum observed denominator is 1.0016 and no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric, defined as the maximum of normalized- and physical-action NRMSE, is 2.414×10^{-6} against a frozen 10^{-3} gate. This primary certificate concerns pole-free denominators and floating-point fidelity to the deployed rational operator. It does not claim uniform agreement with exact RMS scaling or measure matched alternative-normalizer runtime overhead.

The reconstruction ladder extends from individual attention heads through FFNs and per-block replay to a separate sequential learned-forward certificate. That final record starts at exact prepared model tensors and reaches the linear action head on 48 fixed inputs. It is numerical and input-conditioned, not a compact symbolic contraction or an input-general full-policy proof. The ViT source-partition audit remains layerwise and descriptive. Figure 4 reports the frozen numerical gates and the depth profile without assigning causal importance.

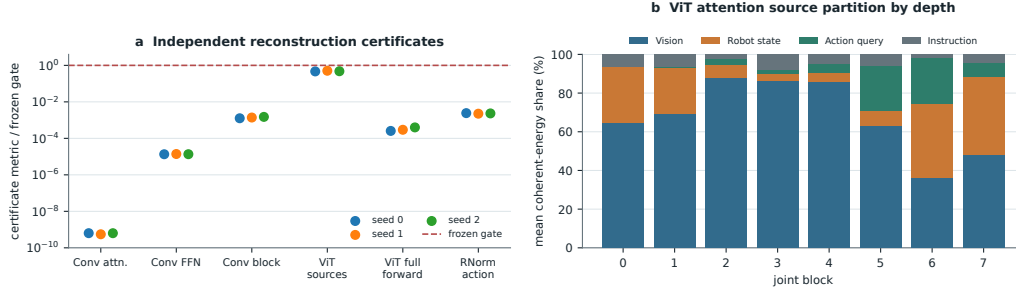


Figure 4: **Frozen reconstruction ladder and descriptive ViT source partition.** Panel (a) reports observed-to-gate ratios for independent Conv attention, joint-FFN, and complete joint-block reconstruction, the ViT source-partition identity, the sequential ViT learned-forward action replay, and primary RationalNorm action fidelity. All plotted primary numerical gates pass. Panel (b) shows descriptive mean coherent-energy shares after the attention output projection across 384 checkpoint-input cases per joint block. The depth trend is not a causal importance or grounding result. The learned-forward replay excludes preprocessing, fixed action unnormalization, chunk execution, simulator dynamics, and success.

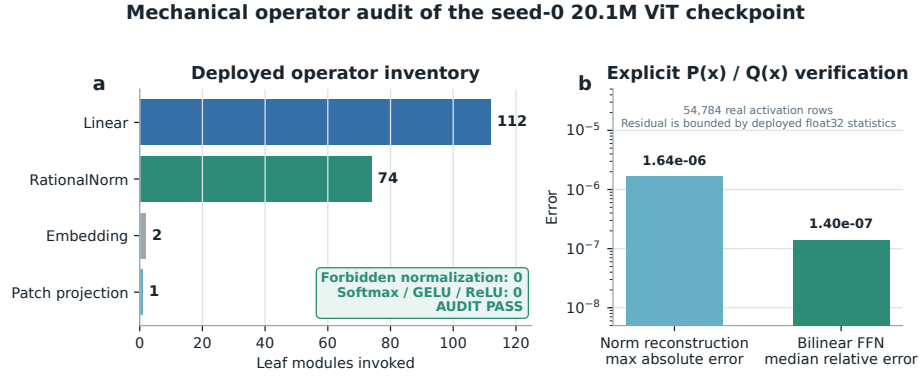


Figure 5: **Mechanical audit of the seed-0 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.